

MIANEH, Iran, Islamic Republic of

WMO: 407160

Lat: 37.450N

Lon: 47.700E

Elev: 3642

StdP: 12.86

Time Zone: 3.50 (IRN)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.9	17.1	4.7	8.0	16.0	9.8	10.4	21.3	20.8	50.5	18.3	45.6	1.2	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.5	101.9	65.2	98.9	64.6	96.3	64.4	68.6	92.7	67.3	91.4	66.2	90.6	9.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.6	90.5	78.9	58.8	84.9	79.0	57.4	80.4	77.6	35.1	93.0	33.9	90.8	33.0	91.0	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	19.3	16.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			6.2	71.5	7.6	2.7	0.7	73.4	-3.7	75.0	-8.0	76.5	-13.5	78.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.8	32.2	37.2	48.5	57.1	67.5	78.1	83.6	82.4	74.0	60.7	46.1	36.5		
	DBStd	18.92	8.76	8.44	7.63	6.56	5.15	4.85	4.74	4.35	4.90	6.78	6.87	6.74		
	HDD50	1618	551	359	118	16	0	0	0	0	0	6	149	419		
	HDD65	4207	1016	777	514	248	33	0	0	0	3	167	566	883		
	CDD50	4825	0	1	70	229	543	844	1043	1004	720	337	33	1		
	CDD65	1940	0	0	1	11	111	394	578	539	273	33	0	0		
	CDH74	22817	0	0	26	169	1170	4443	7211	6582	2847	369	0	0		
	CDH80	12028	0	0	2	38	360	2302	4225	3764	1262	75	0	0		
	WSAvg	5.7	2.8	3.4	5.9	5.7	6.5	8.4	9.0	8.7	7.0	5.0	2.5	3.1		
	PrecAvg	11.4	1.2	1.0	1.4	1.9	1.7	0.4	0.2	0.1	0.2	0.8	1.3	1.1		
(15) Precipitation	PrecMax	16.6	3.1	2.4	3.6	3.8	2.9	1.1	1.3	0.8	1.5	2.9	4.4	3.1		
	PrecMin	7.2	0.2	0.2	0.4	0.3	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.4	0.6	0.5	0.8	0.9	0.7	0.3	0.3	0.2	0.3	0.7	1.1	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.4	61.1	77.5	84.5	90.9	102.1	105.8	104.1	97.0	86.1	68.3	58.3		
		MCWB	43.0	46.1	51.6	55.7	60.0	65.9	65.9	64.7	63.5	58.6	51.1	45.1		
	2%	DB	51.1	56.5	71.1	78.7	87.4	97.5	103.1	100.8	93.6	82.1	64.3	53.2		
		MCWB	41.3	44.6	50.7	55.5	59.1	63.7	65.8	64.2	62.6	57.3	50.2	43.1		
	5%	DB	47.7	52.7	66.4	74.3	84.1	94.4	100.0	98.5	90.6	78.5	60.6	49.4		
		MCWB	39.7	42.9	48.4	54.3	58.3	63.1	65.1	64.5	62.1	56.2	48.7	41.7		
	10%	DB	43.9	49.4	62.3	70.4	80.8	91.3	96.8	95.9	87.6	74.5	57.4	46.2		
		MCWB	38.0	41.4	47.1	52.5	57.8	62.0	65.2	64.3	61.4	55.2	47.8	40.1		
	0.4%	WB	45.9	48.9	53.9	58.8	66.7	71.4	70.1	69.6	67.1	62.1	54.9	46.7		
		MCDB	51.9	57.2	71.7	72.4	79.8	93.7	96.8	93.4	88.6	79.7	62.0	54.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.2	45.8	51.7	57.1	62.2	67.2	68.6	68.1	65.5	59.4	52.4	44.4		
		MCDB	48.6	53.4	68.2	72.3	77.7	90.9	94.4	91.4	87.5	75.7	59.4	49.9		
	5%	WB	40.3	43.9	49.5	55.6	60.4	65.2	67.4	66.9	64.1	57.7	50.3	42.6		
		MCDB	46.4	50.9	63.8	70.5	77.8	89.1	92.5	90.5	85.5	73.1	57.8	48.0		
	10%	WB	38.4	42.1	47.8	54.0	59.2	63.5	66.4	65.8	62.7	56.3	48.6	40.9		
		MCDB	43.9	48.5	60.5	67.7	76.8	86.9	91.7	89.8	83.2	70.7	56.4	45.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.4	17.4	21.1	22.4	23.5	25.8	25.5	26.3	26.5	22.4	18.6	16.1		
		MCDBR	20.8	23.1	27.9	28.5	27.5	28.3	28.1	28.8	29.6	29.1	24.5	20.4		
		MCWBR	12.5	13.0	12.5	12.1	9.1	9.4	8.7	9.5	11.0	12.0	12.8	12.2		
		MCDBR	17.9	19.5	24.3	23.9	24.3	26.3	25.6	26.2	26.4	24.8	20.1	17.3		
	5% WB	MCWBR	11.2	11.2	11.4	10.9	9.3	9.8	8.3	8.8	9.7	10.7	10.4	10.6		
(40) Clear-Sky Solar Irradiance	taub		0.327	0.366	0.427	0.508	0.503	0.482	0.524	0.482	0.423	0.417	0.350	0.321		
	taud		2.276	2.160	2.007	1.843	1.894	1.969	1.887	1.985	2.121	2.080	2.255	2.316		
	Ebn at Noon		268	270	262	247	252	256	244	252	261	247	256	262		
	Edh at Noon		31	40	51	64	62	58	62	55	45	43	32	28		
(44) All-Sky Solar Radiation	RadAvg		797	1087	1361	1657	2079	2498	2446	2248	1837	1260	844	673		
	RadStd		68	115	131	150	113	130	63	61	82	96	110	91		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (11 neighbors)	+0.59	N/A	N/A	+1.75	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page